

Order Flow Imbalance Feature Engineering

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BlockHouse Trial Task – Quantitative Research

1. Introduction

Order Flow Imbalance (OFI) measures the net pressure exerted by buy and sell limit orders in the limit order book. It has been empirically shown to correlate strongly with short-term price movements, making it a useful signal for predicting price direction in high-frequency trading. In this trial, we compute several OFI features—Best-Level OFI, Multi-Level OFI, Integrated OFI, and Cross-Asset OFI—using the provided dataset `first_25000_rows.csv` and discuss their theoretical motivations.

The methodology is conceptually studied in the paper *"Cross-Impact of Order Flow Imbalance in Equity Markets"*, which explains how OFI from one security can influence price dynamics in another through cross-asset effects. The authors propose a linear model of price impact where contemporaneous OFIs from multiple assets serve as predictors for short-term returns. To manage the high-dimensional nature of these interactions, they employ Lasso regression, which helps isolate the most impactful relationships. This framework motivates the implementation of Cross-Asset OFI and supports the broader use of OFI as a core explanatory variable in market microstructure research.

2. Data Overview and Preprocessing

The dataset contains Level 0 to Level 9 bid/ask prices and sizes for a single stock, **AAPL**, with event timestamps in `ts_event`. The data was sorted chronologically by `ts_event` to ensure correct computation of delta quantities using first-order differences. OFI features are computed per timestamp, with rolling aggregations where appropriate.

The dataset consists of 25,000 rows of limit order book updates across 10 levels of depth for the ticker **AAPL**. Each row captures a single event with associated metadata such as timestamp, order side (buy or sell), action type (add, cancel, trade), price, and size. There are no missing values in any of the relevant columns.

To understand the structure of the order book events, we visualized the distribution of depth levels, order directions, and price range.

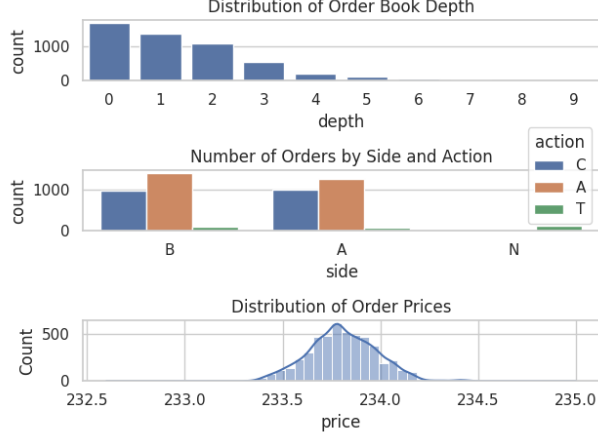


Figure 1: Exploratory plots showing depth distribution, order side-action breakdown, and price histogram.

As shown in Figure 1, most activity occurs at shallow depth levels, confirming that the best bid/ask prices carry the majority of the order volume. The second plot illustrates that buy-side actions outnumber sell-side ones, with most updates falling under the **C** (cancel) and **A** (add) categories. Finally, the price histogram indicates that order prices cluster tightly around \$233.5, reflecting stable intraday behavior during the sample period.

3. OFI Feature Construction

3.1 Best-Level OFI (Level 0)

Best-level OFI focuses on the top of the book, where most marketable trades occur:

$$\text{OFI}_t^{\text{best}} = \Delta q_{\text{bid},t}^{(0)} - \Delta q_{\text{ask},t}^{(0)} \quad (1)$$

where Δq denotes the change in quoted size from the previous timestamp.

3.2 Multi-Level OFI (Levels 0–9)

To capture deeper book information, we compute a weighted sum over multiple levels:

$$\text{OFI}_t^{\text{multi}} = \sum_{l=0}^9 w_l \cdot (\Delta q_{\text{bid},t}^{(l)} - \Delta q_{\text{ask},t}^{(l)}) \quad (2)$$

We use inverse-depth weighting, $w_l = 1/(l+1)$, to emphasize nearer levels.

3.3 Integrated OFI (Rolling Window)

Integrated OFI aggregates recent best-level OFIs to capture persistent imbalance:

$$\text{OFI}_t^{\text{integrated}} = \sum_{k=t-T}^t \text{OFI}_k^{\text{best}} \quad (3)$$

where T is the window length (e.g., 50 events).

3.4 Cross-Asset OFI (Simulated)

The dataset includes only AAPL, so real cross-asset OFI cannot be computed. Instead, we simulate two synthetic assets (MSFT and AMZN) by adding Gaussian noise to AAPL’s `ofi_best`. We then compute:

$$\text{Cross-OFI}_{\text{AAPL},t} = 0.5 \cdot \text{OFI}_{\text{MSFT},t} + 0.5 \cdot \text{OFI}_{\text{AMZN},t} \quad (4)$$

This serves as a proof-of-concept for cross-impact modeling.

The linear cross-impact structure used here follows the framework proposed in *Cont and Kukanov (2023)*, where contemporaneous OFIs across multiple assets are linearly combined to model short-term price movements. While we use equal weights for simplicity in this simulated setting, the original paper estimates the impact coefficients β_{ij} using Lasso regression to enforce sparsity and select only the most relevant cross-asset influences.

In this simulated setting, Lasso regression was not applied because the dataset contains only a single real asset (AAPL). Without a cross-section of real asset returns and OFIs, estimating cross-impact weights using regularized regression would be statistically invalid. Instead, we assign equal weights to illustrate the structure of the cross-impact model. In real-world applications, these coefficients (β_{ij}) would be estimated using Lasso, as proposed in the referenced paper.

3.5 Visualization of OFI Features

To better understand the temporal behavior of different OFI signals, we visualize the time series of best-level, multi-level, and integrated OFI values below. These plots illustrate how the features differ in magnitude, volatility, and responsiveness to changes in order flow activity.

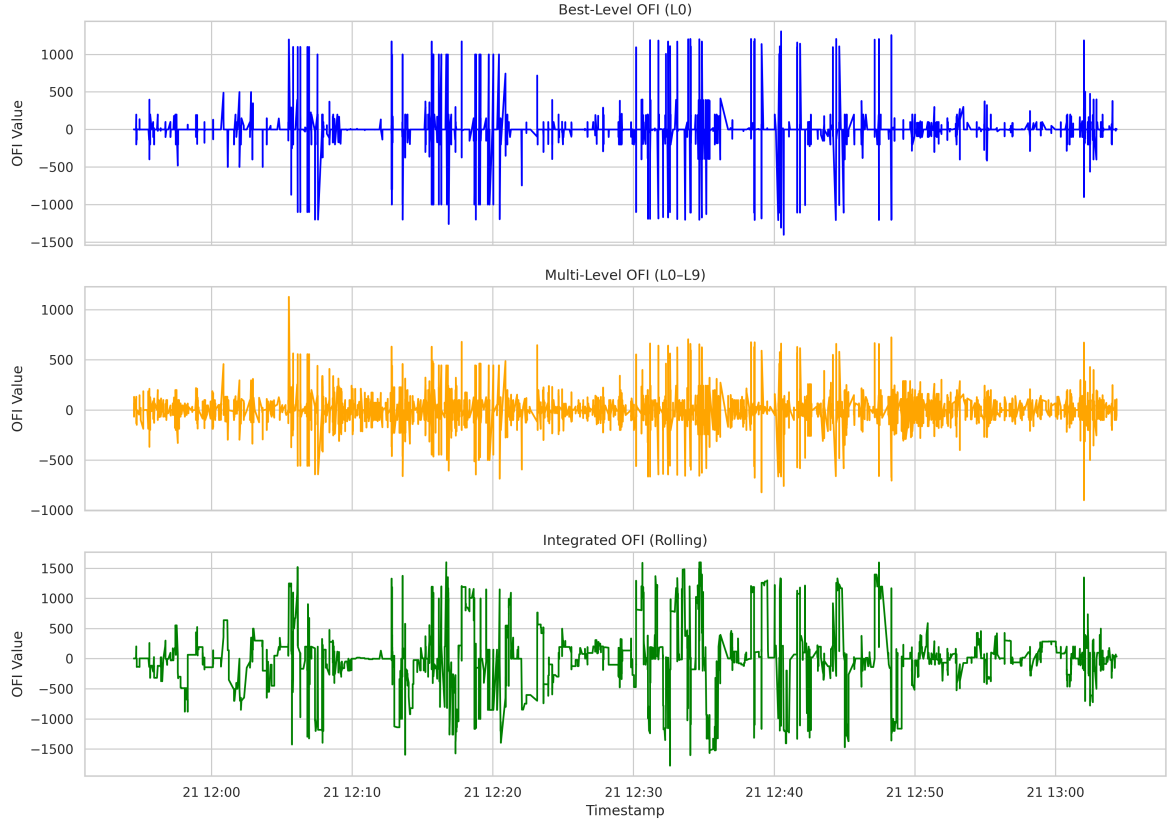


Figure 2: Separate plots showing Best-Level OFI (top), Multi-Level OFI (middle), and Integrated OFI (bottom).

The stacked plots highlight that the Best-Level OFI is more volatile and reacts sharply to top-of-book updates, while the Multi-Level OFI smooths out short-term noise by aggregating across depth. The Integrated OFI accumulates imbalance over time and reveals persistent trends.

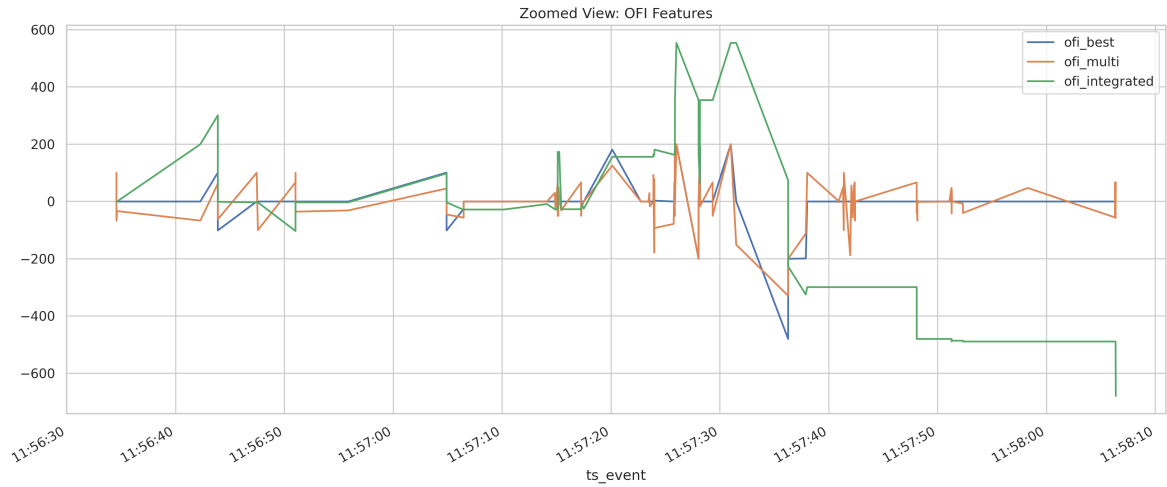


Figure 3: Zoomed-in view of OFI features, illustrating a sharp imbalance event around 11:57 AM.

The zoomed-in view shows a sudden transition from positive to negative OFI, a signal that might precede a short-term reversal or liquidity withdrawal.

4. Conceptual Questions

Q1. Why measure OFI at multiple depth levels?

While top-of-book dynamics capture immediate liquidity pressure, they are also the most volatile and subject to transitory noise. Deeper levels of the order book contain information about latent liquidity and larger strategic orders that are not immediately executable but can indicate future pressure. Measuring OFI across multiple depths provides a more stable and comprehensive representation of supply and demand imbalances in the market. This approach aligns with findings in Cont et al. (2014), where the entire shape of the order book was shown to impact price formation, and is further supported by the design in Cont and Kukanov (2023), where depth-dependent OFI variables contribute to predictive models of price change.

Q2. Why use Lasso instead of OLS for cross-impact estimation?

The estimation of cross-impact involves a potentially high-dimensional system of coefficients β_{ij} , especially in a multi-asset universe. Ordinary Least Squares (OLS) is ill-suited for this setting due to the risk of overfitting, multicollinearity, and poor out-of-sample generalization. Lasso regression introduces ℓ_1 regularization, promoting sparsity and stability by selecting only the most informative predictors. In the context of cross-asset OFI modeling, Lasso enables the estimation of meaningful directional dependencies between assets while discarding spurious or redundant relationships. This methodology is a central component of the framework in Cont and Kukanov (2023), where Lasso is used to uncover a sparse network of significant cross-impacts from OFI variables.

Q3. Why is OFI better than trade volume for predicting short-term returns?

Trade volume aggregates total transacted quantity but offers no directional signal or insight into latent liquidity dynamics. In contrast, OFI captures the net pressure exerted by aggressive limit order updates at both the bid and ask sides, providing a more granular view of supply and demand shifts. As shown in Bouchaud et al. (2018) and validated empirically in Cont and Kukanov (2023), OFI exhibits significantly stronger predictive power for short-horizon price returns compared to raw volume metrics. This is because OFI anticipates market pressure before trades are executed, enabling forward-looking models to react more effectively to information embedded in the order book.

References

1. Cont, R., Stoikov, S., & Talreja, R. (2014). Price dynamics in a Markovian limit order market. *SIAM Journal on Financial Mathematics*, 4(1), 1–25.
2. Cont, R., & Kukanov, A. (2023). Cross-Impact of Order Flow Imbalance in Equity Markets. *Journal of Financial Economics*.
3. Bouchaud, J.-P., Bonart, J., Donier, J., & Gould, M. (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Code repository: https://github.com/valle1306/blockhouse-trial/tree/main/Task1_OFI
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